Algorithm: Bus Route Management System

Step 1: Start.

Step 2: Initialize the circular doubly linked list, stack, and queue.

Step 3: Insert four default bus stops: Sangamner, Akole, Shirdi, and Nashik.

Step 4: Display the main menu and accept the user's choice.

Step 5: If the choice is 1, add a new bus stop and push its name onto the stack.

Step 6: If the choice is 2, search for the specified bus stop and delete it if found.

Step 7: If the choice is 3, search for a bus stop and display its position.

Step 8: If the choice is 4, display the bus route in the forward direction.

Step 9: If the choice is 5, display the bus route in the backward direction.

Step 10: If the choice is 6, display the current, previous, and next stops.

Step 11: If the choice is 7, undo the last recorded addition using the stack.

Step 12: If the choice is 8, open Queue Management to add, delete, or process pending stops.

Step 13: If the choice is 9, free the remaining linked-list nodes and exit the program.

Step 14: If the choice is invalid, display an error message.

Step 15: Repeat the menu until the user selects Exit.

Step 16: Stop.